

Additional Challenges

43

- More flexible bitrate allocations
 - The presented work only considers a few quality levels
- What is a good quality (QoE) metrics?
 - Subjects can rotate their heads anyway **they** like...
 - Content may be very different from one another: **Tourism** versus **extreme sports**; **single** or **multiple** objects?
- How to guide viewers? Compare against the traditional movie director...
- Video stabilization and stitching ← images are from multiple cameras
- Current 360 videos only support 3 Degree-of-Freedom (DoF), how to support 3DoF+ and 6DoF?
- Can we support multiple focal planes (e.g., with lightfield videos) ← important for AR glasses
- **Anything else? I guess content is the key!**

Take-Away Messages

- 360° video technology is an interesting area with many new challenges.
- Fundamentally 360° videos require a tremendous amount of resources (e.g., bandwidth, storage).
- Some of the existing techniques can be adapted, for example MPEG-DASH, but many parts need to be optimized for a high quality user experience.
- Much interesting work still needs to be done!
- Also, new (more exciting) content types, such as **point clouds** and **lightfield** images are around the corner!

Questions

chsu@cs.nthu.edu.tw

